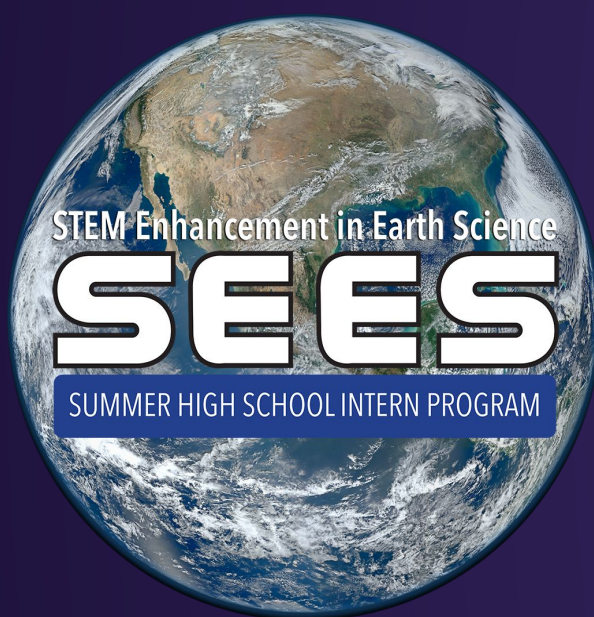
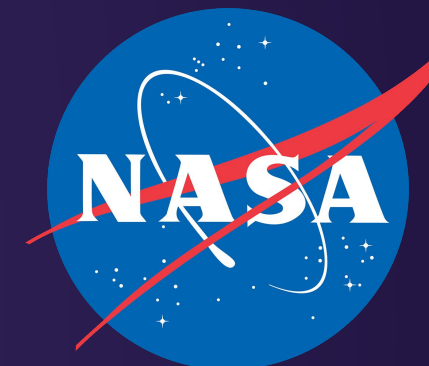




REFINING THE AGE AND DISTANCE TO OPEN CLUSTER NGC 2194



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Abstract

Open clusters serve as natural laboratories for studying stellar evolution and galactic structure, as their stars share a common origin, age, and composition. We present a photometric analysis of the open cluster NGC 2194 to refine its age and distance. Six images of the cluster, taken with B, V, and R filters at two exposure times (3 s and 120 s) using the 0.6 m Southeastern Association for Research in Astronomy (SARA) telescope in Cerro Tololo, Chile, were analyzed. Broadband B and V filter photometry was performed in AstrolmageJ to catalog stars, measure apparent magnitudes, and classify spectral types. Our dataset of 816 stars significantly expands upon the 120 stars examined in Cuffey's 1942 study. Construction of a Hertzsprung–Russell diagram revealed a main-sequence turnoff point near B- to A-type stars. Comparing to stellar evolution models, this corresponds to a maximum age of 400 Myr. After applying reddening corrections and the distance modulus, we derived a refined distance of about 1076 parsecs (~3510 light-years). These results improve upon prior research and provide a framework for future studies of open clusters.

Introduction

Open clusters are loosely bound groups of stars that originate from the same giant molecular cloud. Sharing a common age and chemical composition, they typically contain a few dozen to several thousand stars within a region only a few light-years across.

- Problem:** Lack of comprehensive data on stars within distant open clusters
- Gap:** Limits precise determination of cluster age and distance
- Importance:** Refining these measurements improves stellar evolution models and calibrates galactic distance scales
- Goal:** Determine the age and distance to open cluster NGC 2194 using photometry

Data

- Images taken with 0.6 m SARA telescope in Cerro Tololo, Chile
- Advantages of reflecting telescopes: no chromatic aberration, higher light gathering power
- Broadband filters (B, V, R) isolate specific wavelength ranges, improving signal-to-noise ratio and enabling main-sequence fitting
- Dataset consists of six images, two per filter (3 s and 120 s exposures)
- For analysis, only the 120 s B and V images were used in AstrolmageJ



Figure 1. NGC 2194 astronomical image (Rutherford)

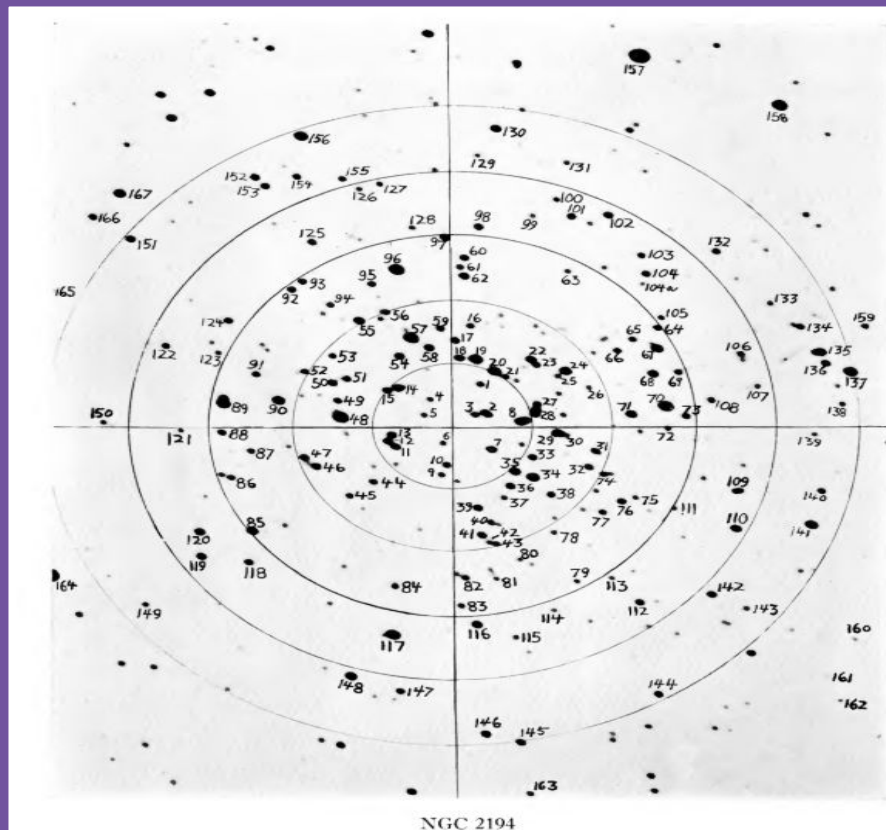


Figure 2. NGC 2194 numbering system (Cuffey)

- Referenced James Cuffey's study "The Galactic Clusters NGC 2126 and NGC 2194,"
- Utilized his star numbering system and age/distance approximations

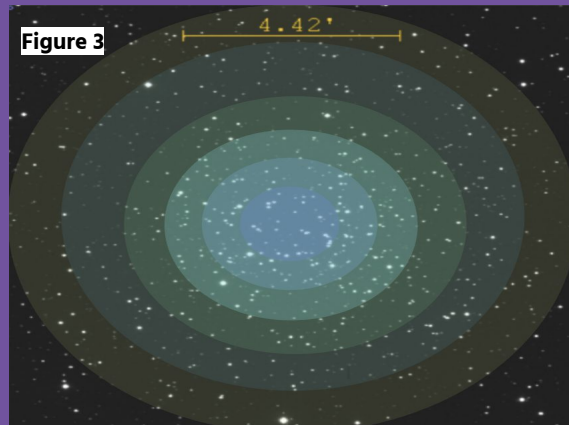
Methodology

1. Star Numbering System

- Developed a system to keep track of all stars for analysis
- Divided cluster into rings and assigned each ring to a team member to number
- Small # = inner, large # = outer

2. Photometry in AstrolmageJ

- Calculated apparent magnitudes of stars
- Used stars with known magnitudes as references for target star magnitudes
- Photon counts within an aperture were converted into brightness values



3. Star Classification

- Recorded all data, including star number, ring number, B and V apparent magnitudes, into a spreadsheet
- Classified stars based on color (B–V) and brightness (V)

Star Ring	V (Green Filter)	B (Blue Filter)	B-V	B-V Corrected (true color index)	V corrected	Star Spectral Class
1 Inner	14.85	15.666	0.816	0.433	13.6244	F
2 Inner	14.905	15.771	0.866	0.483	13.6794	F
3 Inner	14.885	15.768	0.883	0.5	13.6594	G
4 Inner	14.484	15.471	0.987	0.604	13.2584	G
5 Inner	13.415	15.036	1.621	1.238	12.1894	M
6 Inner	14.214	15.081	0.867	0.484	12.9884	F
7 Inner	16.215	17.271	1.056	0.673	14.9894	G
8 Inner	13.553	14.46	0.907	0.524	12.3274	G
9 Inner	13.132	14.745	1.613	1.23	11.9064	M
10 Inner	13.203	14.813	1.61	1.227	11.9774	M

Table 1. Spreadsheet with information for each star

Color Index (B-V)	Spectral Type	Life Span (years)
-0.4	O	$<1 \times 10^6$
-0.2	B	3×10^7
0.2	A	4×10^8
0.5	F	4×10^9
0.7	G	1×10^{10}
1.0	K	6×10^{10}
1.6	M	$>1 \times 10^{11}$

Table 2. Table relating color index, spectral type, and maximum cluster age (University of Washington)

4. Reddening

- Interstellar dust scatters blue light, making stars appear redder
- Measured by comparing observed B–V to the true color index
- Reddening value = 0.383 mag (WEBDA); corrected B–V and V using this

5. H-R Diagram + ZAMS Line

- H-R diagram plots brightness (V) vs color (B–V) to show stellar evolution
- Overlaying ZAMS line identifies stars on the main sequence vs. those evolving off by determination of the turnoff point

6. Age and Distance Calculation

- Age was found by identifying the spectral type of stars at the turnoff point and comparing to stellar evolution models
- Distance (d) was determined using the distance modulus equation, which requires an absolute (M) and apparent magnitude (m)
 - Equation: $d = 10^{(m-M+5)/5}$
- The Sun served as a reference star for the absolute magnitude
- Used a similar G-type star in the cluster for the apparent magnitude

Results and Discussion

Cluster Composition:

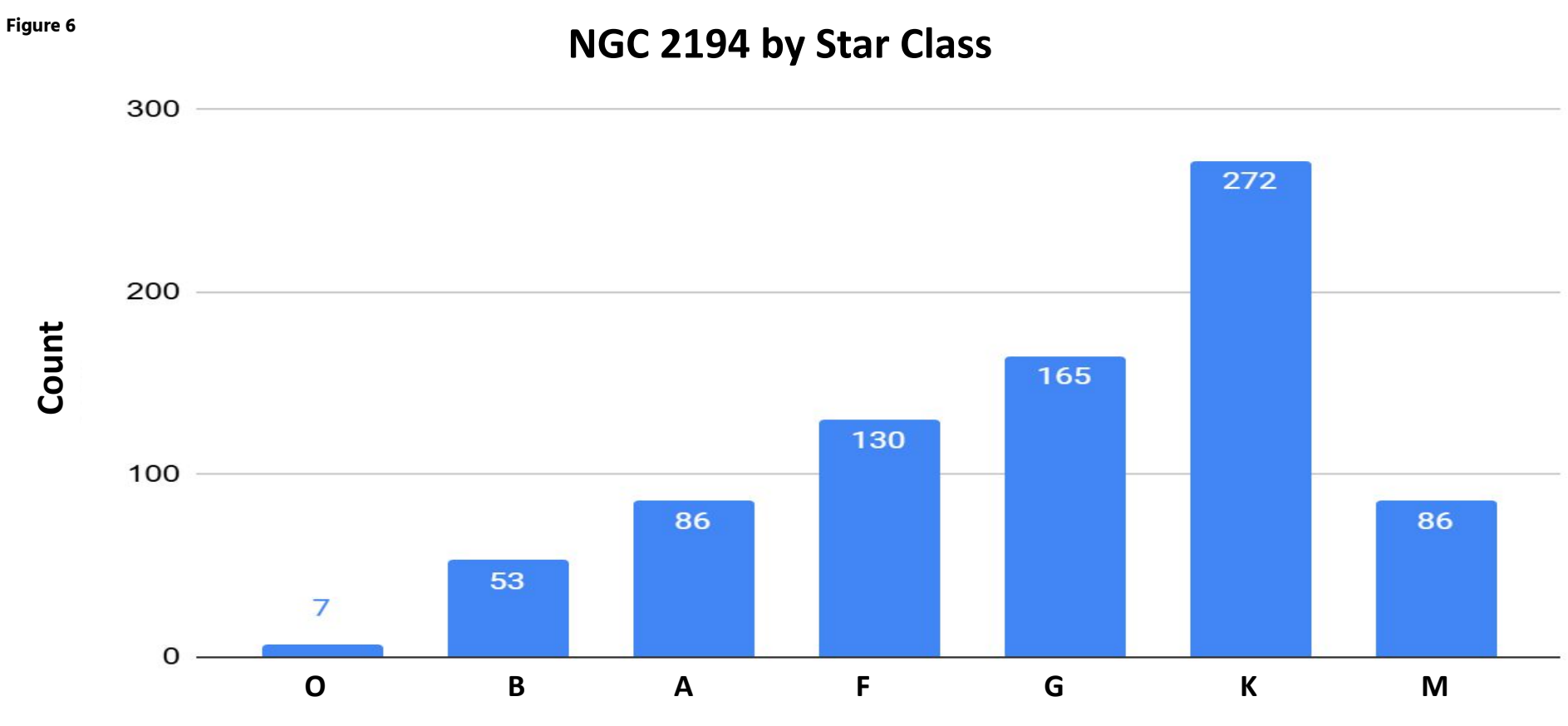
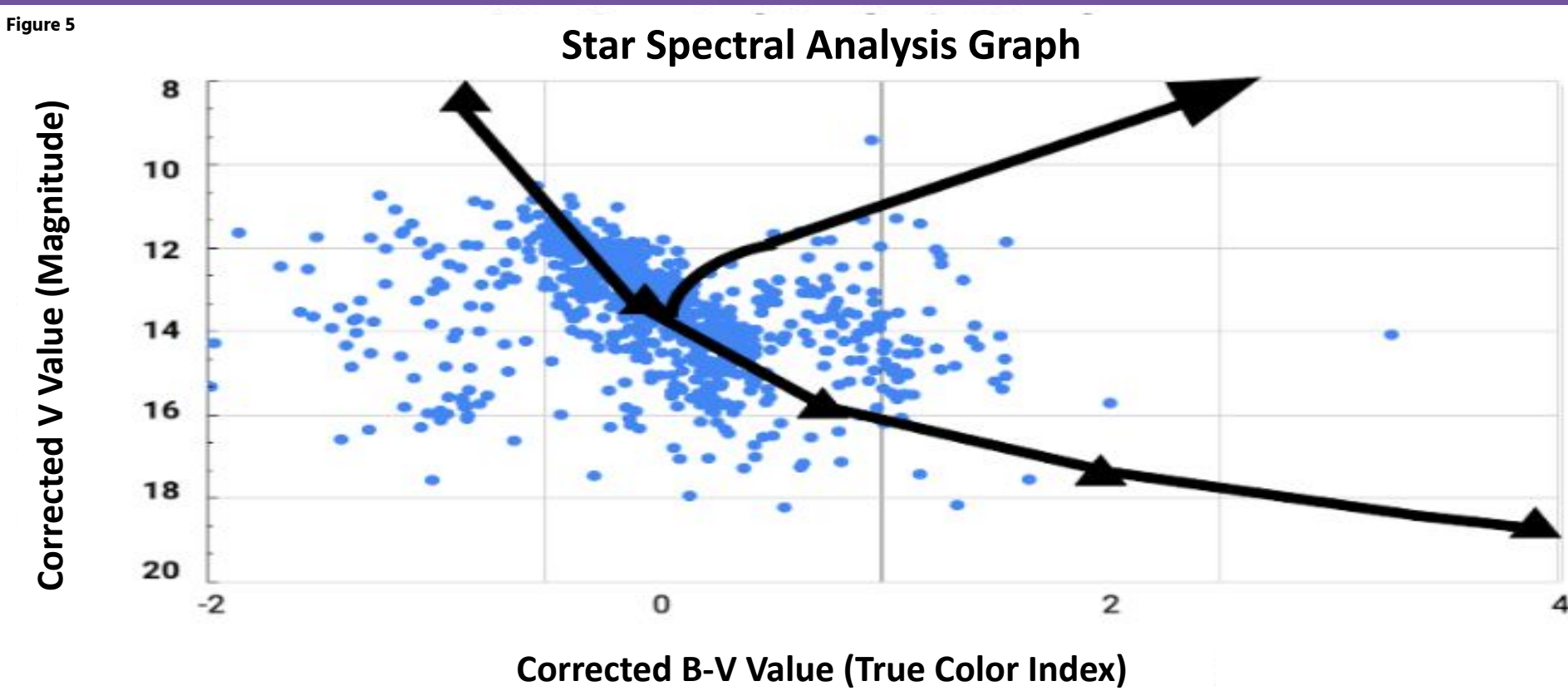
- Inner Ring: dense central cluster with F-, G-, K-, and many M-type stars
- Middle Rings: A- and F-type stars, with some cooler K- and G-type stars
- Outer Rings: fewer stars, mostly colder K- and M-type stars

Cluster Age (H-R Diagram):

- Turnoff point: (B–V = 0, V = 13.5). Corresponds to B- to A- type stars, giving a maximum cluster age of 400 Myrs

Distance Estimate:

- The cluster lies at approximately 1,076 parsecs (~3,510 light years).



- Cuffey's 1940s study identified only 120 stars in NGC 2194 and estimated a cluster distance of 1,200 parsecs
- Our analysis detected 816 stars and a distance of approximately 1,076 parsecs
- The main sequence turnoff star remains consistent as a B- to A- type star

References

- J. Cuffey, *The galactic clusters NGC 2126 and NGC 2194*, Astrophysical Journal, 97, 93–97 (1943).
- University of Washington, *Cluster color–magnitude diagrams and the age of stars*, 2005.
- Masaryk University, *WEBDA Database: Open Cluster NGC 2194*. Available: <http://webda.physics.muni.cz>